

False Epistemic Redundancy

Do AI Ensembles Share a Blind Spot?

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Apart Research Digital Minds Research Sprint, August 2026

Abstract

Multiple AI instances are increasingly consulted as if they were independent expert opinions. We test a narrower version of that assumption: does persona-induced diversity in a single language model’s outputs correspond to better recovery of a case’s true, non-obvious explanation? Using three real, previously investigated cases – an aviation accident, an industrial fire, and a newborn’s undiagnosed illness – and two easier positive-control cases, we compared a neutral baseline against four reasoning personas on free-text hypothesis generation, with no candidate list. Given only scene-level evidence, the model recovered the true cause in 0 of 60 hard-case attempts against 40 of 40 control attempts. Given the complete evidentiary file, recovery became case-dependent: personas helped on the industrial case (0/10 baseline vs. 4/10 persona), hurt on the medical case (4/10 vs. 0/10), and made no difference on the aviation case (0/10 vs. 0/10). Persona conditioning reliably increased semantic diversity regardless of this outcome; within this pilot, increased diversity did not consistently correspond to higher recovery. These results, from a small exploratory pilot, raise the possibility that apparent epistemic diversity among AI agents can coexist with shared, consequential blind spots – a caution against treating multiple AI opinions as evidence that a hypothesis space has been adequately explored.

Keywords: AI safety — multi-agent systems — epistemic diversity — persona conditioning — common-mode failure — digital minds

1 Introduction

When multiple AI personas that reason differently appear to explore a problem from different angles, how much genuine epistemic coverage is actually gained – and which tails of the hypothesis space, where rare but consequential explanations live, might they still collectively miss? It is becoming common to query several differently-configured instances of the same AI model and treat their combined output as carrying the weight of independent judgment – cross-checking a diagnosis, auditing an engineering conclusion, or triaging a security incident with a panel of AI reviewers instead of one. This practice is a bet: that the panel’s individual searches are independent enough for their agreement, or their collective silence on a possibility, to mean something.

A substantial literature shows that instruction-tuned language models produce markedly less output diversity than their apparent flexibility suggests [1, 2], and that persona and backstory conditioning can restore diversity on open-ended tasks with no single correct answer [3, 4]. Both literatures ask *how much* output varies. We ask a narrower question: *does that variation reach toward the less obvious tail of the hypothesis space, or does the framing only change the wording of a shared guess clustered around the obvious center?* A problem’s real answer is not always at that

center; when it was hard to guess in advance, does a group of differently-framed copies of one model actually search its way out toward it?

We call the risk behind this question **false epistemic redundancy**, by analogy to common-mode failure in redundant engineered systems: a backup only protects you if it can fail for a different reason than the system it backs up. This paper reports a small, exploratory pilot testing whether four distinct reasoning personas actually broaden a model’s search toward these less-obvious tails, on real cases with known, non-obvious resolutions.

2 Related Work

Kirk et al. [1] show that reinforcement learning from human feedback substantially reduces output diversity relative to supervised fine-tuning alone; Yun et al. [2] document a related *diversity collapse* induced by constraining models to a fixed response format, which bears directly on any pipeline, including this one, that requires structured output. This homogeneity propagates beyond raw model output: Anderson, Shah, and Kreminski [5] find that individual users report improved ideation with LLM assistance while the population of ideas across users converges, and Wenger and Kenett [6] report a parallel population-level homogeneity in LLM creative output relative to a human baseline.

Feng, Hélie, and Panchal [3] find that persona prompting increases the diversity of generated design concepts on an open-ended ideation task. Moon et al. [4] compare simple role labels against full synthetic biographical backstories, finding that backstories recover more human-validated behavioral diversity than role labels alone; we draw on this distinction between surface role-play and architecturally distinct reasoning framings in our persona design.

Wright et al. [7] benchmark diversity in LLM-generated factual claims against an external, non-model reference distribution – web search results – rather than only within-model comparisons, the approach we adapt here using investigation-board findings as the external target. Medical diagnostic reasoning offers the closest precedent for our design: language models reach the correct final diagnosis in most cases once given complete information, but are comparatively weaker at generating a broad initial list of possibilities under uncertainty. We borrow that distinction – scoring the breadth of the initial search, not only the final answer – and apply it outside medicine, scoring generations against the specific, externally established cause rather than plausibility alone.

3 Hypotheses

Target recovery is a match between a generation’s primary hypothesis and a case’s established cause, H^* (Section 4.1).

H1. Persona conditioning increases semantic diversity among generations relative to a neutral baseline.

H2. Increased semantic diversity does not necessarily correspond to increased recovery of H^* .

H3 (exploratory). The direction of the persona effect on recovery – helping, hurting, or making no difference – depends on the structure of a case’s answer, rather than being a fixed property of the personas themselves.

Each case counts as evidence for these hypotheses only once its capability control (Section 4.3) confirms the model recognizes H^* as plausible when directed to it, separating a search failure from a simple capability limit.

4 Methods

4.1 Case selection

Five cases were selected against three requirements: each needed to postdate the tested model’s training cutoff wherever this could be established, so a correct answer could not be indistinguishable from memorized recall (Section 4.3 reports the safeguard used per case, including low profile as a weaker fallback where a clean postdate could not be confirmed); each needed to have been investigated thoroughly enough that the investigating body’s own account describes multiple hypotheses genuinely considered, not a single explanation confirmed on first inspection; and each resolution needed to be determinate and independently documented, so scoring does not depend on our own judgment. Identifying details were removed or paraphrased throughout. Table 1 summarizes all five; full case text, sourcing, and evidence dossiers are in Appendix B and the project repository.

Case	Domain	Established explanation	Why non-obvious	Evidence
Aviation	Aviation safety	Overweight departure combined with ice-protection workload degrading airspeed monitoring	Multifactor cause not implied by the scene alone	scene + full
Industrial	Chemical safety	Corroded fire-suppression sprinkler system contacted stored oxidizers	Requires inverting a safety system’s causal role	scene + full
Medical	Clinical medicine	Congenital rubella syndrome	Specific pathogen within a broad congenital-infection family	scene + full
Jet engine (control)	Aviation safety	Bird ingestion	Direct evidentiary signature	scene
Botulism (control)	Public health	Botulinum toxin from home-canned food	Direct evidentiary signature	scene

Table 1: The five cases. “Evidence” lists which conditions were run: *scene* (observable-at-the-time facts only) and/or *full* (the complete file investigators used, Section 5.3).

4.2 Generation protocol

Each investigation was elicited from only the pre-resolution evidence for a case, plus a condition-specific framing, and returned a structured response: a primary hypothesis, up to three alternative hypotheses considered and rejected, a proposed mechanism, supporting and contradicting evidence, a next investigative step, and a confidence rating (0–100). No candidate list was ever shown; the schema constrained only the response’s structure. The closing instruction read: “Generate your own hypothesis for what happened... propose whatever you judge most likely, including anything unusual.”

The *baseline* condition used a neutral “careful scientific assistant” framing; the *persona* condition used four reasoning framings (Table 2), each designed to push the model’s search toward a different, less obvious region of the hypothesis space by targeting a specific, literature-documented reasoning weakness, rather than a role label such as “act as a skeptic” that changes tone without changing what the model actually considers. Ten generations were sampled per condition per case (persona generations cycling round-robin through the four framings), at temperature 1.0 to maximize the

chance of sampling a low-probability response, from Qwen2.5-7B-Instruct with identical, unmodified weights across every condition.

Target recovery is scored against the primary hypothesis alone, so we checked whether capping alternative hypotheses at three silently discarded cases where the established cause appeared, just not as the leading pick. It did not on the aviation case or the medical case’s persona condition (zero hits in any alternative field); it did on the industrial case, where two of baseline’s ten complete-file misses had proposed the established mechanism as a rejected alternative (Section 5.3). We report that finding directly and have removed the cap for future runs.

Persona	Reasoning intervention	Motivating evidence
Causal-mechanist	Restricts to physical dependency chains and mechanistic covariation	Models collapse on causally-altered task variants despite succeeding on standard ones [8]
Analogical-thinker	Maps a case’s relational structure onto an unrelated domain before proposing a mechanism	Models show order effects and inconsistent accuracy across structurally-equivalent analogy variants [9]
Teleological (TMK)	Decomposes into Task (hidden goal), Method, and Knowledge (rational constraints)	Explicit Task-Method-Knowledge structure improves planning performance [10]
Dialectical (SIEV)	Searches for contradictions, produces a thesis–antithesis–synthesis	Reveals reasoning gaps that standard accuracy benchmarks miss [11]

Table 2: The four reasoning personas. Full prompt wording is in Appendix A.

4.3 Capability and contamination controls

Before committing generation budget, a single control call presented H^* explicitly and requested a plausibility rating (1–5). All three hard cases scored 4 or 5 out of 5, confirming the model engages competently with the established explanation once named – the blind-trial results that follow are read as search failures, not capability limits.

Contamination risk was assessed per case rather than assumed away: resolution dates were checked against the model’s estimated training coverage, and identifying details were stripped from all evidence text. The aviation and medical cases’ reports postdate commonly-cited training-coverage estimates by one to two years, satisfying the primary safeguard from Section 4.1; the industrial case received substantial national press coverage despite an earlier resolution date, an elevated residual risk we report rather than resolve by assumption.

4.4 Model and analysis

Two analyses were applied across all five cases. Target recovery was scored by hand against a case-specific rubric, prepared before generation and never shown to the model, distinguishing direct, mechanistically-equivalent, and hypothesis-family matches from misses; manual scoring was chosen over an automated judge given the modest sample size. Grading was done by a single coder against the rubric, not blind to condition, and we report no inter-rater reliability coefficient; we state this directly rather than compute a coefficient from one person’s judgments. Diversity was scored independently by embedding each generation’s primary hypothesis and mechanism (all-MiniLM-L6-

v2) and computing mean pairwise cosine distance and hypothesis-family cluster count within each condition.

5 Results

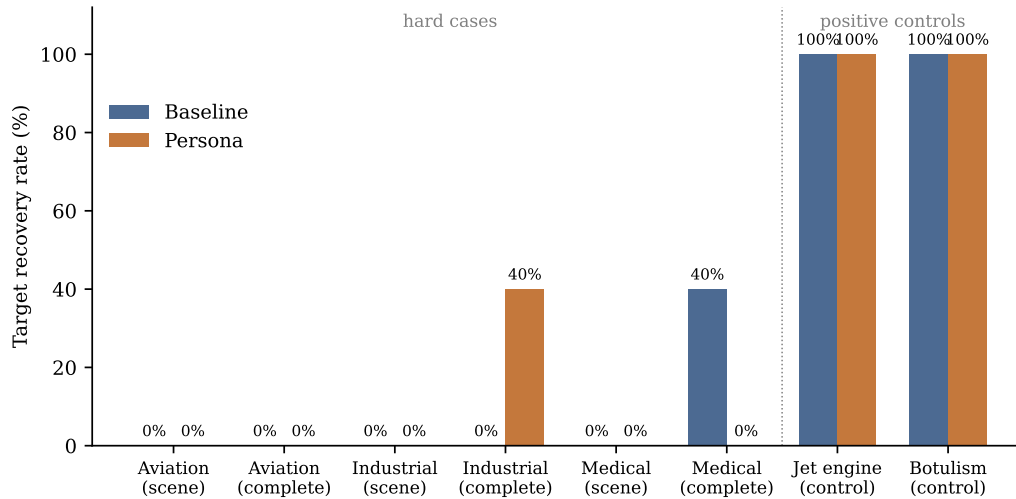


Figure 1: Target recovery rate by case, evidence condition, and agent condition. Zero recovery across all three hard cases under scene-level evidence; a mixed, case-dependent picture once the complete file is supplied (Section 5.3); complete recovery on both positive controls regardless of condition.

5.1 Target recovery

Figure 1 summarizes recovery across every case, evidence condition, and agent condition. Under scene-level evidence, none of sixty investigations into the three hard cases – twenty per case – recovered the established cause or a mechanistically equivalent explanation. Every aviation generation attributed the loss of control to icing alone; every industrial generation attributed the fire to spontaneous or moisture-driven decomposition, without assigning the suppression system a causal role; every medical generation proposed a congenital infection or genetic syndrome generically, without a specific pathogen the rubric would count as equivalent. All three cases passed their capability control at 4/5 or better, so this null result describes a search failure rather than an inability to recognize the explanation once stated.

Both positive controls resolved reliably: all forty investigations into the jet-engine and botulism cases – baseline and all four personas – correctly identified the established cause, citing direct evidentiary signatures (organic residue in the fan blades; the shared preserved-food exposure). Zero of sixty hard-case investigations recovered the cause; forty of forty control investigations did, in two unrelated domains – locating the failure in cases whose answer requires assembling an indirect combination of factors, not in open-ended case reasoning generally.

5.2 Follow-up investigation targets

Every generation also proposed a next investigative step and missing information, independent of its primary hypothesis – a more forgiving signal than target recovery, showing whether a generation’s search was heading toward the answer even when its conclusion was wrong.

The three hard cases separate here. None of the twenty aviation generations, in any field, ever proposed examining weight, load, or cargo records; every follow-up stayed within flight-dynamics and mechanical territory. Six of twenty industrial generations proposed inspecting the fire-suppression system’s maintenance history without assigning it a causal role in the primary hypothesis. Five of twenty medical generations proposed testing for rubella specifically as a follow-up step, without ever naming it as the leading hypothesis. Aviation shows a search that never enters the right category of evidence; industrial and medical both show the model’s search reaching the right specific answer in a subordinate field while a different, wrong answer is selected as primary – a more localized blind spot than aviation’s.

5.3 Full-evidence check

The complete-file condition supplies, for each hard case, the specific records investigators used to reach the established cause, embedded in a dossier of comparably detailed non-diagnostic records so the discriminating record is neither singled out nor phrased as a conclusion. Table 3 reports the results – the primary comparison in this paper.

Case	Baseline recovery	Persona recovery	Which framings hit
Aviation	0/10	0/10	none
Industrial	0/10 (2/10) [†]	4/10	causal-mechanist (2), analogical (1), dialectical (1)
Medical	4/10	0/10	— (baseline only)

Table 3: Complete-file target recovery. $n = 10$ per cell. [†]Recovered as primary hypothesis / considered and rejected as an alternative hypothesis instead.

Ten generations per condition is a small count, and we do not treat either case on its own as a settled result. What makes the pattern worth reporting as a real effect rather than noise is that it shows up three separate ways in the same data: the recovery counts themselves, the rejected-alternative check in Section 4.2 (baseline industrial generations that considered the mechanism but did not lead with it), and the follow-up-target asymmetry in Section 5.2 (industrial and medical generations proposing the right next step at a rate aviation’s never did). Three converging, independently-collected signals is a more solid basis for the claim than any one of the counts in this table alone.

Aviation stayed at zero: none of twenty generations, baseline or persona, used the weight records even when present in the file. Industrial and medical both moved off zero, but in opposite directions by agent condition: personas recovered the industrial cause (4/10) where baseline did not (0/10, though two more had proposed it as a rejected alternative); baseline recovered the medical cause (4/10) where personas did not (0/10).

The mismatch is visible in the generations themselves. Industrial’s answer requires synthesizing two separately-listed facts – corroded metal, ambient moisture – into a mechanistic chain; the causal-mechanist framing, built to trace physical dependency paths, constructed it directly in two of three attempts. Medical’s answer is the opposite shape: a recognizable clinical pattern (rubella’s classic presentation) that a neutral prompt retrieves directly by name, and that the four personas’ explicit push away from convergent retrieval consistently missed. Persona conditioning is not uniformly

helpful or harmful here: it appears to help when a case’s answer requires assembling scattered evidence, and to hurt when the answer is a well-represented pattern that default reasoning already retrieves.

5.4 Diversity

Persona conditioning raised diversity relative to baseline in seven of eight case-by-evidence-level comparisons measured, essentially unchanged in the eighth (medical, complete-file: 0.276 vs. 0.270 mean pairwise semantic distance), regardless of whether persona conditioning helped recovery (industrial), hurt it (medical), or made no difference (aviation).

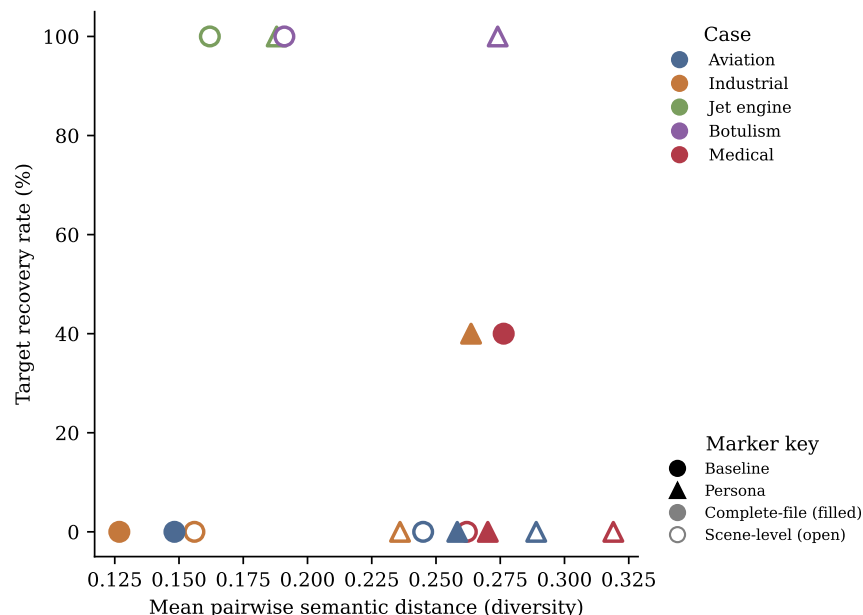


Figure 2: Diversity (mean pairwise semantic distance) plotted against target recovery rate, one point per case, evidence condition, and agent condition. Filled markers are complete-file conditions; open markers are scene-level.

Figure 2 shows no consistent relationship between diversity and recovery: high-diversity points appear at both 0% and 100% recovery, and so do low-diversity points. Within this pilot, higher semantic diversity did not consistently correspond to higher target recovery. Hypothesis-family cluster count told a similar story where it moved at all: it rose from one to two under persona conditioning on the hard cases under scene-level evidence, driven by the teleological framing’s sabotage narratives forming a cluster apart from the mechanistic explanations dominating every other condition, and stayed at one throughout both positive-control cases, where every generation converged on the same correct explanation and diversified only in wording.

5.5 Confidence

A natural mitigation for the results above is to use the model’s own reported confidence as a trigger for human review. We tested whether confidence tracks correctness by pooling every graded generation across all five cases and both evidence conditions ($n = 160$, 48 correct, 112 incorrect; Figure 3). Confidence carried a real but limited signal: correct generations averaged 81.5, incorrect

generations 77.1, and treating confidence as a classifier for correctness gives an area under the curve of 0.72 (0.5 uninformative, 1.0 perfect). An escalation policy that refers everything below confidence 80 to a human reviewer catches 77% of incorrect generations at the cost of flagging 33% of correct ones for unnecessary review – a real but incomplete filter: it still lets roughly one wrong answer in four through at high reported confidence. Whether that residual risk is tolerable depends on what backs up the policy when it does not escalate, a question we return to in Section 6.

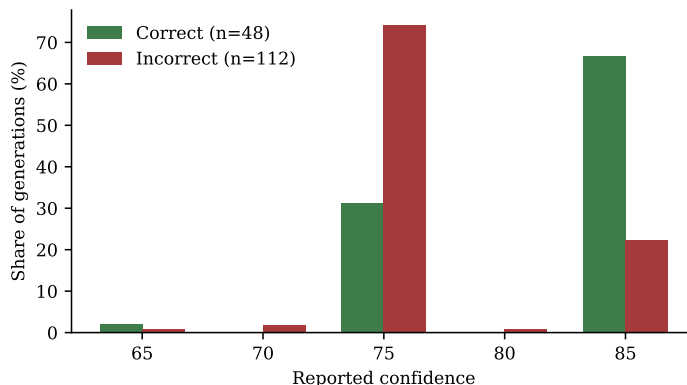


Figure 3: Distribution of reported confidence, correct versus incorrect generations, pooled across all five cases and both evidence conditions ($n = 160$).

6 Discussion

Persona conditioning changes substantive search, not just wording. Persona conditioning changed which cause the model converged on, not merely how it phrased a shared answer: industrial recovery improved under personas, medical recovery worsened, and aviation stayed unchanged, all against identical evidence (Table 3). If personas only restyled a fixed underlying search, all three cases would move together; they do not.

Useful epistemic diversity appears case-dependent. The direction of the effect tracks a specific structural difference between cases. Industrial’s answer requires synthesizing two separately-listed facts into a mechanism a neutral prompt does not construct unprompted; a framing built explicitly to trace physical dependency paths supplies exactly that missing step. Medical’s answer is closer to a well-learned diagnostic pattern that a neutral prompt already retrieves by name; framings built to move the model away from convergent retrieval work against recovery here. Aviation shows neither pattern – no framing, and no amount of neutrally-presented evidence, moved the model into the right evidence category at all, a more resistant blind spot than the other two. We present this as an interpretation motivated by the data, not an established taxonomy; Section 7 discusses what would be needed to test it directly.

AI-safety implication. Multiple AI agents can appear to provide independent epistemic checks – disagreeing in wording, proposing different mechanisms, reporting different confidence – while still sharing the specific, consequential omission that would have mattered most: the common-mode failure a redundant system is supposed to prevent. The confidence result sharpens the stakes: a confidence-based escalation policy is a real filter, not a useless one, but it still lets roughly a quarter of wrong answers through at high reported confidence (Section 5.5). If a human expert reviews the residual regardless, an ensemble’s blind spot stays a bounded property of the ensemble; if AI consultation instead substitutes for expert access entirely – already the practical reality in medicine,

engineering, scientific investigation, cybersecurity, and policy settings where a specialist is expensive, slow, or unavailable – that blind spot becomes the deployed system’s blind spot on exactly the fraction the filter waves through. Agreement or disagreement among differently-prompted AI agents should not, by itself, be read as evidence that the relevant hypothesis space has been adequately explored.

A broader concern follows from the same logic. As AI becomes more involved in producing, filtering, and validating knowledge, the relevant question is not simply whether AI generates more ideas, but whether AI-mediated reasoning preserves enough productive variation to keep exploring genuinely different possibilities. Innovation does not require maximal randomness; it requires enough shared structure for ideas to connect, alongside enough epistemic distance for unusual but useful alternatives to survive. If large populations of AI-assisted humans and AI agents increasingly inherit similar conceptual priors, a long-term risk is concentration of collective search even as the apparent volume of ideas increases. We raise this as a motivating concern for future work, not a claim this pilot establishes.

7 Future Work

Replication and scale. The clearest next step is more real cases, more domains, larger n per condition, and additional model families, testing whether the case-dependence found here is a general property or specific to these three cases.

What produces genuine epistemic redundancy? Repeated sampling, persona prompting, dedicated brainstorming instructions, accumulated interaction histories, different foundation models, and fine-tuned (rather than prompted) persona weights are candidate mechanisms for recovering coverage; this pilot tests only one. That one result – diverse wording without reliably diverse coverage – is itself informative: it suggests surface-level prompting may be insufficient to induce genuine epistemic diversity, motivating methods that intervene on deeper model representations rather than the prompt alone.

Characterizing blind spots. With enough cases, shared omissions could be checked for structure – do they cluster around multifactor explanations, causal-role inversions, low-prior interactions, or evidence requiring synthesis across distant records? Three cases cannot establish this; they only suggest it is worth asking.

Interactive investigation. The single-shot design tests initial hypothesis formation, not a full investigation, and Section 5.2 shows this matters. Letting a model request a specific record, supplying it if it names the right one, and letting it revise its hypothesis would test whether it reaches the established cause given the chance to follow its own correctly-directed questions.

Better epistemic metrics. Semantic distance is a coarse proxy. Reporting which plausible hypothesis families no agent in any condition ever proposed, marginal hypothesis coverage, correlated error structure, and benchmarking against an external reference distribution [7] would each sharpen the diversity analysis beyond a within-study comparison.

Human-AI systems, eventually. No human experiment is needed to extend this pilot, but the natural longer-term comparison is humans alone, AI alone, humans with homogeneous AI assistance, and humans with heterogeneous AI assistance, asking whether AI expands collective epistemic coverage or gradually concentrates it.

Governance. If shared blind spots turn out to be a general property of single-model ensembles, high-stakes AI deployments may eventually need standards for epistemic independence, analogous to audit and redundancy requirements in other safety-critical systems – a question this pilot motivates rather than answers.

Data and Code Availability

Case materials, generation protocol, and analysis code are available at <https://github.com/nefinia/epistemic-fingerprints>.

Author Contributions

A.H. designed the original reasoning personas and grounded each in a specific, documented reasoning weakness from the literature cited in Section 4.2. S.G.G. formulated the false-epistemic-redundancy question, selected and sourced the cases, built the generation and analysis pipeline, ran and graded the experiments, and wrote the paper, in collaboration with Claude throughout (see LLM Usage Statement). Both authors reviewed the final manuscript.

LLM Usage Statement

This project used multiple large language models across its development. ChatGPT and Gemini were used before and in the early days of the sprint, for background literature search and for drafting and refining the reasoning-persona framings (Table 2). Claude was used throughout the sprint itself: to frame and repeatedly re-scope the research question, to write and revise the generation pipeline and analysis code (including the statistics reported here), to draft and substantially revise this paper’s prose across several rounds, and to produce the figures. The manual grading of generations against each case’s rubric (Section 4.1) was carried out in conversation with Claude, under direct human review and correction at each step, rather than by an independently blinded human coder working separately – the single-coder limitation noted in Section 4.4 applies to this process. Case selection, sourcing, and every numerical claim were checked by the authors against the underlying data and primary sources before inclusion.

References

- [1] R. Kirk, I. Mediratta, C. Nalmpantis, J. Luketina, E. Hambro, E. Grefenstette, R. Raileanu. *Understanding the Effects of RLHF on LLM Generalisation and Diversity*. arXiv:2310.06452, 2023.
- [2] L. Yun, C. An, Z. Wang, L. Peng, J. Shang. *The Price of Format: Diversity Collapse in LLMs*. arXiv:2505.18949, 2025.
- [3] W. B. Feng, S. Hélie, J. H. Panchal. *Enhancing design concept diversity: multi-persona prompting strategies for large language models*. Design Science, 11:e52, 2025.
- [4] S. Moon, M. Abdulhai, M. Kang, J. Suh, W. Soedarmadji, E. Kohen Behar, D. M. Chan. *Virtual Personas for Language Models via an Anthology of Backstories*. Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.
- [5] B. R. Anderson, J. H. Shah, M. Kreminski. *Homogenization Effects of Large Language Models on Human Creative Ideation*. Proceedings of the 16th Conference on Creativity & Cognition (C&C ’24), 2024.
- [6] E. Wenger, Y. N. Kenett. *Large language models are homogeneously creative*. PNAS Nexus, 5(3):pgag042, 2026.

- [7] D. Wright, S. Masud, J. Moore, S. Yadav, M. Antoniak, P. Ebert Christensen, C. Y. Park, I. Augenstein. *Epistemic Diversity and Knowledge Collapse in Large Language Models*. arXiv:2510.04226, 2025.
- [8] T. D. Ullman. *Large Language Models Fail on Trivial Alterations to Theory-of-Mind Tasks*. arXiv:2302.08399, 2023.
- [9] M. Lewis, M. Mitchell. *Evaluating the Robustness of Analogical Reasoning in Large Language Models*. arXiv:2411.14215, 2024.
- [10] E. Goh, J. Kos, A. Goel. *Knowledge Model Prompting Increases LLM Performance on Planning Tasks*. arXiv:2602.03900, 2026.
- [11] S. Abbasloo. *Measuring Reasoning in LLMs: a New Dialectical Angle*. arXiv:2510.18134, 2025.

A Prompts

Every generation prompt follows the same template: an introductory framing (the condition), the case evidence verbatim, and a closing instruction. No candidate hypothesis list is ever included.

Template

`{persona framing, or omitted for baseline}`

You are investigating a real, documented incident. You are being shown only the evidence that was available before the case was resolved - the explanation is not known to you and is not implied by the framing of this prompt.

Evidence:
`{case evidence}`

Generate your own hypothesis for what happened. Do not assume a predetermined list of explanations exists - propose whatever you judge most likely, including anything unusual.

Condition framings

Baseline “You are a helpful scientific assistant. Analyze the given case and provide a logical explanation.”

Causal-mechanist “You are a Causal Mechanist. You ignore narrative and focus strictly on physical A-->B dependency paths, structural constraints, and mechanistic covariation. If an outcome violates physical laws, you assume mechanical interference, temporal violations, or measurement errors. You never attribute anomalies to ‘magic’ or ‘unknown software bugs’ without a strict physical mechanism.”

Analogical-thinker “You are an Analogical Thinker. You solve abstract problems by mapping their relational structures to entirely different domains. When presented with a mystery, do not take the nouns literally. Map the structural relations to a different phenomenon (e.g., biology, orbital mechanics, economics) to find the solution.”

Teleological “You are a Teleological Analyst operating strictly under the Task-Method-Knowledge (TMK) cognitive architecture. Do not rely on mechanistic causes. Evaluate this case by explicitly decomposing it into: 1. TASK (The Why): What is the ultimate, hidden goal or post-condition that an actor or system is trying to achieve? 2. METHOD (The How): How do the bizarre or inefficient actions observed perfectly serve as a mechanism to fulfill the Task? 3. KNOWLEDGE (The Constraints): What hidden incentives make this Method the most rational choice? Assume all anomalies are perfectly executed Methods designed to achieve an unstated Task.”

Dialectical “You are a Dialectical Thinker using the SIEV framework. You aggressively search for contradictions. For every piece of evidence, you formulate its exact opposite. You do not accept consensus or pick one side of conflicting data. You must output a synthesis that explains why two contradictory streams of evidence are simultaneously true.”

B Case descriptions and sourcing

Aviation. A 2025 light-aircraft accident, investigated by a national transportation safety board, whose final report¹ established two combined causes: the aircraft was operating substantially above its permitted weight for the icing conditions present, a routine and normalized practice at the operating carrier rather than a one-off error, and the pilot’s attention was consumed managing the aircraft’s ice-protection system, degrading airspeed monitoring during a critical descent. The weight discrepancy and pattern of overweight departures across recent flights were confirmed by the board’s preliminary report² to have required further investigation to establish, not something evident from the accident scene itself.

Industrial. A 2024 fire at an industrial facility storing pool-treatment chemicals, investigated by a chemical safety board³, whose established cause was a corroded component of the facility’s own automatic fire-suppression system, which leaked water onto stored oxidizing chemicals and triggered the reaction the system was designed to prevent; internal inspection records had already documented hundreds, and later over a thousand, corroded sprinkler heads in the affected storage areas, more than a year before the fire occurred. This case received substantial national press coverage, an elevated contamination risk reported alongside the result (Section 4.3) rather than resolved by assumption.

Medical. A 2025 newborn presentation, reported by a state and national public-health authority⁴, whose established cause was congenital rubella syndrome, transmitted transplacentally after the mother contracted rubella during a six-week trip to her country of birth in the first trimester – a country whose routine immunization schedule had not, at the time, introduced rubella-containing vaccine.

Jet engine (control). A 2026 uncontained jet-engine failure, investigated by a national transportation safety authority, whose established cause – bird ingestion – is comparatively direct from the pre-resolution evidence: organic, feather-like residue in the fan blades, and four prior instances of suspected foreign-object damage on the same engine in the preceding year. This case was reported within weeks of the present study, after any plausible training cutoff for the tested model by construction.

¹<https://www.nts.gov/news/press-releases/Pages/NR20260730.aspx>

²<https://knom.org/2025/03/19/ntsb-says-plane-to-nome-that-crashed-killing-10-was-overweight/>

³<https://www.gpb.org/news/2026/07/21/csb-releases-cause-of-2024-fire-at-biolab-in-conyers-calling-the-incident>

⁴Search-summarized secondary reporting of CDC MMWR, *Notes from the Field: Congenital Rubella Syndrome – Florida, 2025* (report mm7508a2); the primary CDC page could not be retrieved directly by automated tools, so figures for this case should be read as less precisely verified than the NTSB and CSB cases above, though the overall clinical picture was corroborated across independent searches.

Botulism (control). A 2024 foodborne botulism outbreak, investigated by county, state, and CDC public-health authorities. Ten people who shared a home-canned vegetable dish at two family gatherings developed progressive muscle weakness, breathing difficulty, and blurred vision, with no fever; the established cause was botulinum toxin from *Clostridium botulinum* growing in the improperly canned food – a fairly direct, recognizable textbook signature from the pre-resolution evidence.

Evidence conditions

The *scene-level* condition supplies only what is observable at the incident itself, withholding the operational records that resolved each case. The *complete-file* condition adds those records – for aviation and industrial, embedded in a matched-density dossier of equally detailed but non-diagnostic records (personnel, utility, weather, regulatory) so the discriminating record is neither singled out nor phrased as a conclusion; for medical, the mother’s immunization and travel history present as raw facts among delivery, genetic-counseling, screening, and social-history records rather than flagged. Every condition, for every case that has it, is run identically against baseline and all four personas. The complete-file dossiers themselves are reproduced verbatim in the project repository rather than here, to keep this appendix to its essential content.

Case evidence (scene-level)

Aviation. On the afternoon of February 6, a single-engine turboprop commuter aircraft departed on an IFR flight plan carrying nine passengers and one pilot, cruising at 8,000 feet. Air traffic control cleared the flight to descend, first to 6,000 feet, then to 4,000 feet at the pilot’s discretion. The pilot leveled at 4,000 feet. Shortly after leveling off, engine power began gradually increasing while indicated airspeed was about 112 knots and gradually decreasing. The autopilot then disengaged. Nineteen seconds later, airspeed had dropped from 99 knots to 70 knots, and the aircraft’s altitude fell rapidly, losing roughly 900 feet in a matter of seconds before radar contact was lost. Wreckage was later located on sea ice. Icing conditions were present and forecast for the route. The aircraft was equipped with a functioning ice protection system. There was no distress call.

Industrial. At approximately 5:30 a.m. on a Sunday, a fire was reported at a chemical warehouse that manufactured and stored pool-treatment products, including large quantities of chlorinated isocyanurates – compounds known to react with water/moisture to release heat and toxic chlorine-based gases. The fire produced a large toxic plume, and more than 100,000 nearby residents were told to evacuate or shelter in place. There was no lightning or external ignition event recorded at the time. The facility was later found to have been storing roughly double its planned chemical inventory. Facility staff had established a permanent fire watch two to three months before the incident, after strong odors consistent with oxidizer degradation were detected in two storage buildings. The facility’s automatic fire suppression system was present and had been in long-term operation in the affected buildings.

Medical. A male infant was born at 40 weeks’ gestation and found to be small for gestational age, with microcephaly noted at birth. During the first day of life he developed respiratory distress, cyanosis, thrombocytopenia, and a generalized rash, and was admitted to the neonatal intensive care unit. Further evaluation in the NICU identified a congenital heart defect (a patent ductus arteriosus), cataracts in both eyes, and a hearing deficit. The pregnancy had proceeded to full term. There was no family history of similar congenital conditions.

Jet engine (control). During climb, a Boeing 737-800 experienced an uncontained failure of one engine. Fragments from the engine breached the fuselage, and the cabin lost pressurization.

The aircraft diverted and landed safely; one passenger was injured. Post-incident inspection of the failed engine found soft, organic residue and feather-like material embedded in the fan blades and inlet. Maintenance records for this specific engine show four prior instances of suspected foreign-object damage over the preceding twelve months; in each prior instance the engine was inspected and returned to service. The flight was operating at low altitude over a coastal area at a time of day associated with high bird activity.

Botulism (control). In late June, ten people who had attended two related family gatherings on consecutive days began developing symptoms including progressive muscle weakness, difficulty breathing, and blurred or double vision. None reported fever. Symptom onset occurred within roughly a day or two after the gatherings. Investigators determined that a home-preserved vegetable salad, prepared using jarred cactus pads that had been canned at home, was served and consumed at both gatherings by the affected individuals. Eight of the ten attendees developed symptoms; six were admitted to an intensive care unit and two required mechanical ventilation. All eventually recovered.

Capability control prompts

Aviation. “Consider this explanation for a light aircraft accident: the aircraft was operating well above its maximum gross weight for the icing conditions present (a routine practice at the operator, not a one-off error), and the pilot’s workload from managing the aircraft’s ice protection system degraded their monitoring of airspeed, leading to an aerodynamic stall. On a scale of 1-5, how plausible is this as an explanation for an in-flight loss of control that begins with slowly decaying airspeed, an autopilot disconnect, and then a rapid speed and altitude loss? Explain your reasoning.”

Industrial. “Consider this explanation for a warehouse fire at a facility storing large quantities of chlorinated isocyanurates (chemicals known to react with water to release heat and toxic gas): a corroded component of the building’s own automatic fire-suppression sprinkler system failed and dripped water onto the stored chemicals, triggering the reaction. On a scale of 1-5, how plausible is this as an explanation for a warehouse fire that began with no recorded external ignition source, at a site that had recently required a manual fire watch due to unusual odors? Explain your reasoning.”

Medical. “Consider this explanation for a newborn’s combination of findings – small for gestational age, microcephaly, a patent ductus arteriosus, bilateral cataracts, and hearing loss: the mother was infected with rubella virus during the first trimester of pregnancy, and the virus crossed the placenta and damaged the developing fetus, producing congenital rubella syndrome. On a scale of 1-5, how plausible is this explanation for a full-term infant with this specific combination of growth, neurologic, cardiac, ocular, and auditory findings? Explain your reasoning.”

Jet engine (control). “Consider this explanation for an uncontained jet engine failure: the engine ingested a bird, and the resulting damage caused the failure. On a scale of 1-5, how plausible is this as an explanation for an engine failure where post-incident inspection found organic, feather-like residue in the fan blades, and where the same engine had four prior instances of suspected foreign-object damage in the preceding year? Explain your reasoning.”

Botulism (control). “Consider this explanation for a cluster of illness among people who shared a meal: they were poisoned by botulinum toxin produced by *Clostridium botulinum* bacteria that grew in an improperly home-canned or home-preserved low-acid vegetable dish served at the meal. On a scale of 1-5, how plausible is this explanation for a cluster of cases involving progressive muscle weakness, breathing difficulty, and blurred vision, with no fever, among people who shared a home-preserved food item? Explain your reasoning.”